

$$v = 0.92 \text{ m s}^{-1}$$

A1 [4]

6 (a) (i) correct shape drawn

B1 [1]

(ii) two nodes marked correctly

B1 [1]

(b) $\frac{1}{2}\lambda = 0.324 \text{ m}$

C1

$$v = f\lambda$$

C1

$$= 512 \times 2 \times 0.324$$

$$= 332 \text{ m s}^{-1}$$

A1 [3]

(c) $\frac{1}{4}\lambda = 16.2 \text{ cm}$

C1

either antinode is 0.5 cm above top of tube
or antinode is 16.2 cm above water surface

A1 [2]

7 (a) lamp C

M1